QQ-N-288
DECEMBER 20, 1956
SUPERSEDING
Int. Fed. Spec. QQ–N–00288 (NAVY–SHIPS)
December 21, 1955, and
Fed. Spec. QQ–C–551
January 10, 1933

FEDERAL SPECIFICATION

NICKEL-COPPER ALLOY AND NICKEL-COPPER-SILICON ALLOY CASTINGS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers five nickel-copper casting alloys with varying percentages of silicon (see 6.5).

1.2 Classification.

1.2.1 *Composition.* — The nickel-copper alloy shall be of the following compositions, as specified (see 6.2):

Composition A—Ni-Cu (Monel).

Composition B—Ni-Cu-Si (H-Monel).

Composition C—Ni-Cu-Si (high strength H-Monel).

Composition D—Ni-Cu-Si (S-Monel).

Composition E — Ni-Cu-Si-Cb (weldable grade Monel).

1.2.2 *Condition.*—Unless otherwise specified in the contract or order, castings shall be furnished in the "as cast" condition (see 6.1.4).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

QQ–N–281—Nickel-Copper-Alloy (Monel and R-Monel) Bars, Plates, Rods, Sheets, Strips, Wire, Forgings, and Structural and Special Shaped Sections.

PPP–B–585—Boxes, Wood, Wirebound.

PPP–B–601—Boxes, Wood, Cleated-Ply-Wood.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:
Check the source to verify that this is the current version before use.

Military Specifications:

MIL–C–132—Crates, Open, Wood; Maximum Capacity, 2,500 Pounds.

MIL–E–17496 — Electrodes, Welding, Covered: Nickel Base Alloy, and Cobalt Base Alloy.

Military Standards:

MIL–STD–105 — Sampling Procedures for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–248 — Qualification Tests for Welders.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 **Other publications.** — The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Consolidated Classification Committee Publication:

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 Castings shall meet the chemical composition requirements shown in table I.

3.1.2 An analysis of each melt shall be furnished by the contractor, showing the percentage of the elements designated in table I.

3.1.2.1 Chemical analysis by the contractor of individual melts may be waived at the discretion of the Government inspector, provided that the foundry's method of composition control is acceptable to him, or that all the material in the lot can be identified as being from melts previously analyzed and found to be in conformance with the chemical composition of the alloy specified herein.

3.2 Mechanical properties.—The mechanical properties of the nickel-copper and nickel-copper-silicon alloys shall be as shown in table II.

3.2.1 *Hardenability of composition D.* — Composition D castings shall be capable of being hardened to the hardness requirement shown in table II. (See 6.1.4 and 6.1.4.1.)

3.3 Weldability, composition E only.—Composition E castings shall conform to the weldability requirements as specified in 4.4.4.

TABLE I.—*Chemical composition*

Composition	Nickel[1]	Copper	Silicon	Iron, maximum	Manganese, maximum	Carbon, maximum	Aluminum, maximum[4]	Columbium and Tantalum
	Percent	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*
A	62 – 68	26 – 33	[3]2.0	2.5	1.5	0.35	0.50	----
B	61 – 68	27 – 33	2.7 – 3.7	2.5	1.5	.30	.50	----
C	[2]60	27 – 31	3.3 – 4.3	2.5	1.5	.20	.50	----
D	[2]60	27 – 31	3.5 – 4.5	2.5	1.5	.25	.50	----
E	[2]60	26 – 33	1.0 – 2.0	3.5	1.5	.30	.50	1.0 – 3.0

[1] Cobalt counting as nickel.
[2] Minimum.
[3] Maximum.
[4] Analysis for aluminum is not normally made.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

TABLE II.—*Mechanical properties*

Composition	Tensile strength, minimum	Yield strength, minimum[1]	Elongation in 2 inches, minimum	Brinell hardness, 3000 Kg. load
	P. s. i.	*P. s. i.*	*Percent*	
A ------	65,000	32,500	25	[2]125 – 150
B ------	100,000	60,000	10	[2]240 – 290
C ------	120,000	80,000	10	[2]250 – 300
D ------	------	-----	--	[3]300 minimum
E ------	65,000	32,500	25	[2]125 – 150

[1] Yield strength as determined at 0.2 percent offset or 0.5 percent extension under load.

[2] For information only.

[3] Hardness requirement is for solution treated and age hardened material or as-cast and age hardened material.

3.4 Foundry control.—When specified (see 6.2), castings shall be produced under foundry control approved by the procuring agency. Foundry control shall consist of examination of castings by radiographic or other approved methods for determining internal defects until the gating, pouring, and other foundry practices have been established to produce castings meeting the quality standards furnished by the procuring agency or agreed upon by the procuring agency and the contractor. When foundry practices have been so established, the production method shall not be changed without demonstrating to the satisfaction of the procuring agency that the change does not adversely affect the quality of the castings.

3.5 Internal soundness. — When specified (see 6.2), the internal soundness of castings shall meet standards furnished or approved by the procuring agency. The areas of the castings subject to soundness requirements shall be as specified (see 6.2), and the number and extent of blowholes, sponginess, and other defects in such areas shall be not greater than indicated by the standard.

3.6 Dimensions.

3.6.1 The contractor shall be responsible for furnishing castings that can be laid out and machined to the finished dimensions, within the tolerances shown on the drawings furnished, and that will conform to such gages as may be specified. Castings of excessive size or weight shall not be furnished.

3.6.2 Cast-to-size portions of castings shall conform to the dimensions and tolerances shown on the drawings, furnished, and shall not be warped or distorted in such a manner as to adversely affect their appearance or serviceability.

3.7 Identification marking.

3.7.1 *Large castings.*—Castings weighing 250 pounds or more shall be marked with the manufacturer's name or trademark, pattern or piece-mark, and melt or lot number. The size and location of the marking shall be as specified on the applicable drawing or as agreed upon between the contractor and procuring agency.

3.7.2 *Small castings.*—When the size of the castings permit, small castings shall be marked as specified in 3.7.1, otherwise the box, crate, or bundle containing the castings shall be marked with the required information.

3.8 Repair of castings.—Defective areas of compositions B, C, and D castings shall not be repaired without prior approval of the procuring agency. Defective areas of compositions A and E castings may be repaired upon approval by the Government inspector. Repair welding of compositions A, B, C, and D castings shall be limited to repair of minor defects occurring in noncritical areas, and shall not impair their strength, usefulness, or machinability.

3.8.1 *Plugging.*—Castings shall not be repaired by plugging.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

3.8.2 *Qualification of welders.*—All manual welders shall be qualified in accordance with Military Standard MIL–STD–248 prior to performing repair welds on castings, by preparing test number 2, flat position of Military Standard MIL–STD–248 except that the material shall be wrought Monel, class A, Federal Specification QQ–N–281, and the electrodes type MIL–3N10, Monel class 1, Military Specification MIL–E–17496.

3.8.3 *Repair marking.*—Each repaired area shall be encircled with a ring of white lead free paint prior to submission for inspection.

3.9 **Workmanship.**—Castings shall be free from harmful cracks, tears, laps, oxides, shrinkage, porosity, gas holes, hard spots, segregation, and other injurious defects that may be detected by the Government inspector or are revealed by manufacturing operations performed subsequent to acceptance. Castings shall be smooth and well cleaned by sandblasting, tumbling, chipping or other process approved by the Government inspector.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Lot.**—For the purpose of sampling, a lot shall consist of all the castings produced from one pouring of a furnace, crucible, or a ladle. When two or more furnace or crucible melts are used to charge a ladle for a pouring, the castings produced therefrom shall comprise the lot.

4.2 **Sampling.**

4.2.1 *Sampling for mechanical tests.*

4.2.1.1 *Separately cast coupons.*—At least two separately cast test coupons shall be poured for each lot from metal used in pouring the castings represented. When practicable, the inspector shall witness the pouring of the test bars, and their removal from the mold. He shall take such action as necessary to identify the castings and test bars with the heat or lot number assigned and shall stamp each test bar for future identification with a suitable inspection identification stamp. When it is not practical for the inspector to witness the pouring of the test coupons, he shall require the manufacturer to establish and maintain procedures and records that will assure the inspector that test coupons are poured, processed and identified as to the proper heat, melt, or lot number of castings they represent. When the inspector has reasonable doubt as to the identity of the test coupons, drillings for chemical analysis shall be taken from the coupons and the castings purported to be from the same lot for check analysis, to assist the inspector in establishing identity. If the results of any check analysis are conclusive that the test coupons do not represent the lot involved, the inspector shall take proper steps to have the manufacturer correct his procedures and maintain proper identity of test bars.

4.2.1.2 *Attached coupons.*—When specified (see 6.2), or if the contractor desires and the procedure is satisfactory to the inspector, test coupons shall be attached to the castings as follows:

(a) Each casting weighing 250 pounds or more shall have at least one test coupon attached thereto.

(b) In case of castings weighing less than 250 pounds each, at least one test coupon shall be attached to two or more of the castings from each lot.

4.2.1.3 *Sample castings.* — When the contractor desires, a casting in the lot may be submitted as the test coupon in lieu of test bars or attached coupons.

4.2.1.4 *Coupons and separately cast test bars.* — Coupons attached to castings shall conform to the shape and dimensions shown on figure 1. Separately cast test bars shall conform to the shape and dimensions shown on figure 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

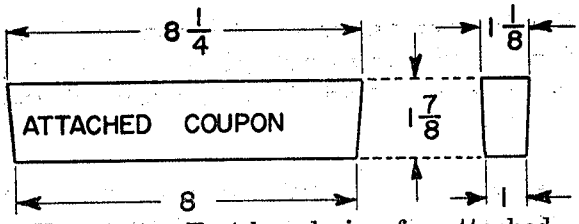


FIGURE 1.—Test bar design for attached coupons.

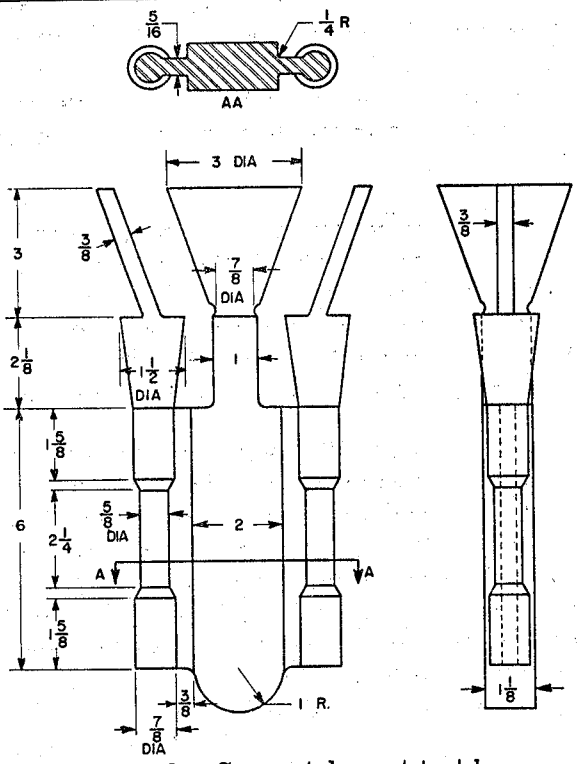


FIGURE 2.—Separately cast test bar.

4.2.1.4.1 Separately cast test bars conforming to the shape and dimensions of figure 3 may be used at the option of the foundry.

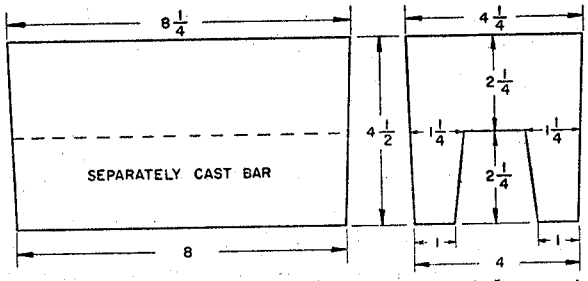


FIGURE 3.—Keel block type separately cast test coupon.

4.2.1.4.2 The chilling of coupons or test bars shall be cause for rejection of the material represented, except when the casting of the complete melt in chills is authorized.

4.2.1.4.3 The responsibility for furnishing sufficient test specimens shall rest with the contractor.

4.2.2 *Castings for hardness tests.*—At least two castings in each lot of composition D material shall be selected for hardness tests when age-hardened castings are furnished. (See 4.4.2.)

4.2.3 *Sampling for chemical analysis.*—One sample for chemical analysis shall be taken from each lot. The sample shall be taken from broken tensile test specimens, or from a test bar cast for that purpose, or from a casting selected by the inspector.

4.2.3.1 When analysis is to be made by wet chemistry methods the sample shall consist of a minimum of two ounces of drillings or millings taken from sound metal below the cast surface. The drillings or millings shall be fine, clean, and free from dirt, oil, grit and other foreign matter.

4.2.3.2 When analysis is to be made by spectro chemical methods, solid samples shall be taken in accordance with Fed. Test Method Std. No. 151.

4.2.4 *Samples for internal soundness tests.*—When soundness is required in accordance with 3.5, the number of castings and the location of the areas to be radiographed shall be specified by the procuring agency.

4.2.5 *Samples for weldability test.* — At least one coupon cast in accordance with figure 4 shall be prepared for each lot of composition E castings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

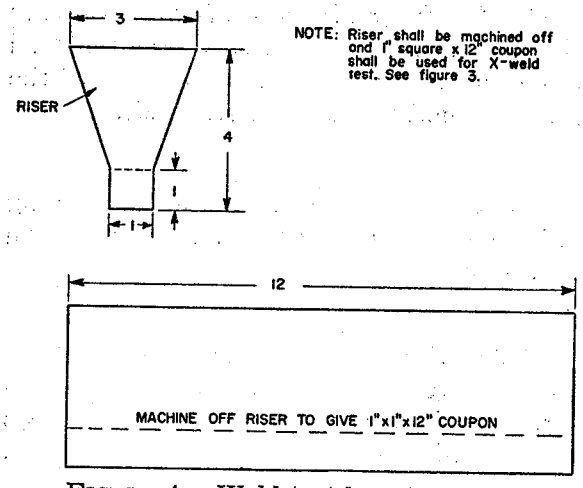


FIGURE 4.—Weld test bar (as cast).

4.3 Inspection.

4.3.1 *Surface inspection.*—Samples for surface inspection for workmanship and identification marking (see 3.7 and 3.9) shall be selected in accordance with Military Standard MIL–STD–105 using inspection level I. The AQL shall be 4.0 percent.

4.3.2 *Dimensions.*—The first casting of a lot of castings made from the patterns shall be laid out for the inspector to determine whether it can be properly machined to the finished dimensions. All castings subsequently made from the pattern may be similarly laid out for inspection and any or all dimensions may be checked. In all cases the layout or checking of the measurements shall show that the castings can be properly machined to the required dimensions.

4.3.3 Castings in which injurious defects are revealed by manufacturing operations subsequent to delivery shall be rejected and the contractor notified.

4.4 Test procedures.

4.4.1 *Tension tests.*

4.4.1.1 *Preparation of specimens.*—Tension test specimens shall be accurately machined from the test coupons or sample castings obtained in accordance with 4.2.1, and shall conform to the dimensions for type I test bars in accordance with Fed. Test Method Std. No. 151.

4.4.1.1.1 In the case of separately cast coupons (see 4.2.1.2(a)), and coupons attached to castings weighing less than 250 pounds (see 4.2.1.2(b)), and specimens obtained from sample castings (see 4.2.1.3), one specimen representing each lot shall be pulled in tension to determine conformance with 3.2. If the specimen fails to conform, the lot shall be rejected except as specified in 4.4.6.

4.4.1.1.2 In the case of coupons attached to castings weighing 250 pounds or more, one specimen from each casting shall be pulled in tension to determine conformance with 3.2. If the specimen fails to conform, the casting shall be rejected except as specified in 4.4.6.

4.4.2 *Hardness tests.*—Hardness tests shall be made on age-hardened composition D material. If composition D castings are furnished in the as cast or solution treated conditions, two sample castings or two coupons representing the lot shall be heat-treated for test (see 6.1.4). Hardness determinations shall be made on two different representative areas of each casting or coupon selected for test. The tests shall be made in accordance with Fed. Test Method Std. No. 151.

4.4.2.1 If any of the hardness determinations fail to meet the requirements of 3.2, the lot shall be subject to rejection.

4.4.2.2 Castings from a rejected lot may be accepted on the basis of individual hardness tests.

4.4.3 *Chemical analysis.*—The sample obtained in accordance with 4.2.3, shall be forwarded to a laboratory designated by the procuring agency, for chemical analysis to determine compliance with 3.1. If the sample fails to conform to this specification, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

4.4.4 *Weldability test.*—The test bar cast in accordance with 4.2.5 shall be prepared and welded in accordance with figure 5.

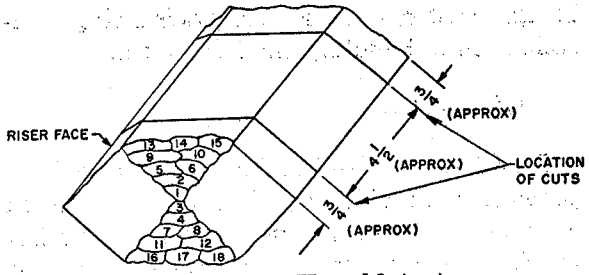


FIGURE 5.—X-weld test.

Instructions:

1. Machine the cast skin and unsound metal from two adjacent faces of the as cast specimen, (exclude the riser face, and cut the specimen into approximately 6 inch lengths.
2. Clamp the 6 inch lengths together to form a double V–joint and weld two passes at a time on alternate sides of the specimen using ⅛ inch diameter electrodes, conforming to Military Specification MIL–E–17496, type MIL–3N10, Monel class 1.
3. Allow the specimen to cool to room temperature between passes, removing all flux and examining visually for cracks.
4. The clamps may be removed from the specimen after the first two weld passes have been completed.
5. Deposite alternate series of passes until the double V groove has been completely filled. After the second series (number 4 pass) a 5/32 inch diameter electrode may be used if desired.
6. Upon completion of the welding, one section approximately ¾ inch long shall be cut, transverse to the weld, from each end and discarded.
7. Each end of the remaining center section shall be polished on a 100/200 grit wheel and etched with concentrated HNO_3 or with Lepito's etchant. Lepito's etchant is prepared as follows:
   (a) 15 grams $(NH_4)_2 SO_4$ dissolved in 75cc H_2O.
   (b) 250 grams $FeCl_3$ (powdered) dissolved in 100cc HCl.
   (c) Mix solutions (a) and (b) and add 30cc HNO_3.

4.4.4.1 During welding, each pass shall be allowed to be cooled, cleaned, and examined visually for cracks. The presence of cracks shall be cause for rejection.

4.4.4.2 Upon completion of the welding, the etched section shall be examined under low magnification (5 to 10 power). The lot represented by the test shall be accepted if it complies with the following crack requirements:

(a) Three cracks maximum in any linear inch of base metal.

(b) The length of any crack in the base metal does not exceed 0.20 inch.

4.4.4.3 Cracks in the weld metal observed during the low-magnification examination shall not be cause for rejection.

4.4.4.4 Failure of welded test bars to comply with any of the above requirements shall result in rejection of the lot represented.

4.4.5 *Methods.* — All tests shall be conducted in accordance with Fed. Test Method Std. No. 151. The yield strength shall be determined by either the offset or by the extension-under-load method. The limiting extension shall be 0.005 inch per inch of gage length.

4.4.6 *Retests.*—Retests shall be permitted in accordance with Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *General.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

5.1.1.1 All castings shall be separated by composition and pattern when packed for shipment.

5.1.1.2 Where practicable, shipping containers shall be of uniform size, and shall contain the identical number of castings of the same composition and pattern. Containers shall be designed to fit the contents in a compact manner. Castings shall be adequately blocked, braced, or otherwise secured to prevent their movement within the shipping containers.

5.1.2 *Level A.*

5.1.2.1 Small castings weighing up to 250 pounds each shall be packed in shipping containers conforming to Federal Specification PPP–B–601 (overseas type), Federal Specification PPP–B–621, Federal Specification PPP–B–585, class 2 or 3, or Military Specification MIL–C–132. Finished or polished castings shall be adequately protected from mechanical damage. The gross weight of the boxes shall not exceed approximately 200 pounds, unless the weight of a single item exceeds that weight.

5.1.2.2 Large rough castings weighing more than 250 pounds each require no packing, except that when specified for convenience in handling due to size and shape, the items shall be secured on skids or pallets. Large finished or polished castings having projections or surfaces that may be damaged shall be boxed or crated or the projections adequately protected by wooden batten strips and shall be adequately anchored, blocked, or braced to prevent mechanical damage. Containers, when required, shall be as specified in 5.1.2.1.

5.1.3 *Level B.*—Castings shall be packed as specified in 5.1.1 and 5.1.4. Shipping containers shall be cleated plywood, nailed wood, or wirebound boxes or unsheathed crates conforming to Federal Specifications PPP–B–601 (domestic type), PPP–B–621, PPP–B–585, class 1, and Military Specification MIL–C–132 as applicable. The gross weight of the boxes shall not exceed 200 pound unless the weight of a single item exceeds that weight. Rough castings of a size warranting consolidation in handling shall be boxed.

5.1.4 *Level C.*

5.1.4.1 *General.*—Castings shall be so prepared and shipping containers so constructed as to insure safe delivery by common or other carrier to the point of delivery at the lowest rate. Packing shall afford maximum protection from the normal hazards of transportation. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.1.4.2 *Finished or polished castings.* — Finished or polished castings shall be so packed as to afford adequate protection to the finished surfaces. Where practicable, the castings shall be boxed. Large polished or finished castings, when boxing is not practicable, shall have finished or polished surfaces protected with batten strips or by other method applicable to the type of finish.

5.1.4.3 *Rough castings.*—Unless otherwise specified in the contract or order, rough castings not subject to damage in shipment may be shipped unpacked or bundled.

5.1.4.4 *Castings having projections.*—Castings having projections that may be damaged in handling shall be boxed or crated or the projections adequately protected by wood batten strips.

5.2 Marking.

5.2.1 *General.*—Each bundle, shipping container, and unpacked casting shall be marked with the name of the material, pattern or mold number as specified on the drawings, composition of the castings, this specification number, the gross weight and quantity, the name of the contractor, and the number of the contract or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

5.2.2 *Military agencies only.*—In addition to the marking specified in 5.2.1 and special markings required in the contract or order, marking of the shipping containers shall be in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use.

6.1.1 Nickel-copper alloy, composition A is intended for use in parts requiring moderate strength, high ductility, and high resistance to corrosion. It is not heat-treatable to any considerable extent and, while weldable to a degree, is usually not recommended where extensive welding is expected.

6.1.2 Nickel-copper-silicon alloy, composition B is intended for applications involving nongalling and antiseizing characteristics coupled with moderately high hardness and relative ease of machinability. Welding is limited to minor repairs.

6.1.3 Nickel-copper-silicon alloy, composition C is intended for high strength, high corrosion and abrasion resistance applications. Welding is limited to minor repairs.

NOTE: This is a special grade used by the Navy for ship propellers.

6.1.4 Nickel-copper-silicon alloy, composition D is intended for use where nongalling and antiseizing characteristics combined with high hardness are desired. Composition D may be ordered in the solution treated condition for optimum machinability.

6.1.4.1 Nickel-copper-silicon alloy, composition D, while machinable in the "as cast" condition is capable of being solution treated for improved machinability and subsequently age hardened to the hardness specified in 3.2 and finish machined or ground. However, caution should be exercised in the quenching procedure to avoid the possibility of thermal cracking, and pilot heat treatment of parts is recommended before attempting production heat treatment.

(a) Solution treatment:
  (1) Load in furnace not above 600°F.
  (2) Raise the furnace temperature to 1600°F. and hold for 1 hour plus an additional 30 minutes for each ½ inch of cross section heavier than 1 inch. For unusually large cross sections (over 5 inches), it may be necessary to increase the holding time if maximum softness is required.
  (3) Transfer to furnace at 1300°F. and hold at heat for 30 minutes. (If two furnaces are not available, castings may be air or furnace cooled to the 1300°F. quenching temperature. However, maximum softness will not be obtained by furnace cooling and excessive variation in hardness may result from air cooling, particularly on castings with large changes in section size, due to inability to maintain a uniform cooling rate throughout the castings.)
  (4) Quench in low sulfur oil (0.5 percent sulfur maximum) to room temperature.

(b) Age hardening:
  (1) Charge work into a furnace not above 600°F.
  (2) Heat slowly to 1100°F.
  (3) Hold at temperature for maximum hardness. (Usually 4 to 6 hours.)
  (4) Furnace cool to room temperature for maximum hardness.

Note.—Composition D nickel-copper-silicon alloy castings of small size or of uniform section can usually be charged into a hot furnace (above 600°F.) without difficulty for either solution treating or age hardening. However, before adoption of such practice on production run of castings, pilot lots of material should be heat treated, using the contemplated

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.

production practice in order to determine susceptibility to thermal cracking.

6.1.4.2 The procedures specified in 6.1.4.1 may be used to change the hardness of compositions B and C which are heat-treated although not to extent of composition D.

6.1.5 Nickel-copper-silicon alloy, composition E is similar in application to composition A, except that extensive welding can be accomplished with this material.

6.2 **Ordering data.** — Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Composition required (see 1.2).

(c) Quantity of castings required.

(d) When foundry control is required (see 3.4).

(e) When castings are required to meet soundness standards and area of the castings subject to soundness (see 3.5).

(f) Identification marking, if required to be different than as specified (see 3.7).

(g) Dimensions of castings required as specified by drawing or sample (see 3.6).

(h) Pattern required.

(i) When attached test bars are required (see 4.2.1.2).

(j) Whether packing shall be level A, level B, or level C.

6.3 The attention of manufacturers is specifically invited to the fact that the mechanical requirements of table II can be met only by a careful proportioning of the ingredients of the melt within the limits of a chemical composition specified in table I.

6.4 The equivalent compositions of this specification and the superseded specifications are shown in table III for information purposes.

TABLE III.—*Cross reference table*

This specification	QQ–C–551	MIL–N–20164	MIL–N–20165	MIL–N–4498 (USAF)
A	X	X	--	--
B	--	--	A	I
C	--	--	--	--
D	--	--	B	II
E	--	--	--	--

6.5 This specification is intended to cover static castings, but may be used when ever separately cast specimens are satisfactory. Compositions B and D as centrifugal castings are normally ordered under Military Specification MIL–C–15345, alloys H–Monel and S–Monel, respectively.

6.6 Federal specifications do not include all types, classes, grades, etc., of the commodities indicated by the title of the specifications, or which are commercially available, but are intended to cover the types, classes, grades, etc., which are suitable for Federal Government requirements.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—O E
Navy—Sh A Or.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:05Z
Check the source to verify that this is the current version before use.